

Patients With Personality: Realistic Patient Simulation through Controlled Diversity and Selective Disclosure

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Abstract

Simulating realistic patient interactions is a key requirement to testing clinical applications of LLMs at scale without time-consuming and expensive user studies. However, existing approaches often lack realism and controllability, often oversharing information unprompted, and failing to capture the wide variability of patient behavior. Here, we introduce *PatientsWithPersonality (PWP)*, a patient simulation framework that generates realistic yet diverse virtual patient responses through explicit personality parametrization over a latent patient state. Grounded in HEXACO, a six-dimensional personality space used to quantify and parameterize human behavioral traits, our approach enables fine-grained control over conversational style, cooperativeness, and information disclosure within a unified framework. In a clinician evaluation, PWP is judged nearly as realistic as recorded human actors and clearly ahead of prior simulators, while being flagged as “too informative” far less often. Conditioning on HEXACO axes yields personas whose configured traits are recoverable by both clinicians and an autorater, span a substantially wider behavioral footprint than the closest baseline, and prevent oversharing. Altogether, our framework paves the way for more accurate and informative LLM benchmarking through our realistic and steerable patient simulator.



Project Website



Code

1 Introduction

Benchmarking large language models (LLMs) plays a central role in guiding model selection, tracking progress, and informing deployment decisions, particularly in the high-stakes medical domain where reliable evaluation is a prerequisite for safe clinical use [1, 2]. Popular approaches such as exam-style question-answering benchmarks for medical LLMs are increasingly saturated and only moderately predictive of clinical capability [3–5]. As a result, these static benchmarks are being supplemented by interactive and agentic evaluations that more closely mirror realistic clinical scenarios, closing the knowledge-practice gap [6–8].

A foundational building block in these interactive benchmarks is a *simulated patient*: an LLM-driven agent that a clinician AI model must interact with. Realistic simulated patients are essential to make evaluations automated and scalable and thus allow us to rapidly improve clinician LLMs before verifying their abilities in costly human studies.

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However, the lack of realism and behavioral fidelity of the simulated patients often distorts benchmark performance. This has direct consequences for benchmark validity: in a large randomized study of more than 1,000 participants, Bean et al. [9] found that simulated patients substantially overestimated model performance and correlated only weakly with results from real patients, concluding that simulation-based evaluation misses the interaction failures that matter most in deployment. Two recurring failure modes explain this gap: First, simulators tend to overshare, volunteering information that was not explicitly asked for, and flesh out underspecified facts with plausible but fictitious content [10]. Second, they produce uniform and cooperative patients, lacking the imperfect recall, variable cooperativeness, and personality-driven variation that characterize real patient encounters [11, 12] (cf. Figure 1).

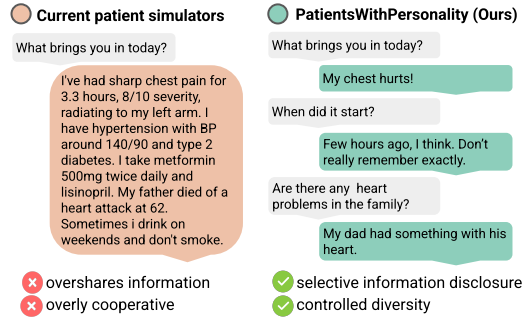


Figure 1: Our *PatientsWithPersonality* framework resolves failure modes of current patient simulators by selectively controlling information disclosure and by allowing to simulate diverse characters grounded in the HEXACO personality space.

Our Contributions. To address both failure modes, we introduce *PatientsWithPersonality (PWP)*, a patient simulation framework that decouples behavioral personality modeling from information disclosure. Specifically, we contribute:

- The *PWP* framework featuring psychometrically grounded personality configuration and dynamic information control through a query-conditioned disclosure mask;
- A conversation simulation protocol that for the first time enables the medical simulator comparison under a single metric set measuring realism, disclosure, and personality fidelity;
- An extensive analysis of multiple simulators showing that *PWP* is more realistic, more diverse, and more controllable than all other tested frameworks.

2 Background & Related Work

LLM-based patient simulators. LLM-driven patient simulators have emerged as a scalable alternative to real humans across a range of clinical applications. They serve as entity in multi-agent benchmarks [7, 8, 13], replace human patient actors in medical education [10, 11, 14, 15], and probe multi-turn communication properties that static tasks cannot capture [16]. The paradigm has further been extended to simulated users for health and lifestyle coaching [17].

Existing clinical designs cluster into two broad families. The first comprises lightweight, prompt-based simulators that are typically components of a larger agentic benchmark rather than research artifacts in their own right. *CRAFT-MD* turns case vignettes into simulated doctor-patient dialogues to evaluate multi-turn reasoning and history taking [8]. *AgentClinic* embeds a prompt-driven patient agent inside a multi-agent, OSCE (Objective Structured Clinical Examination)-like diagnostic benchmark and adds cognitive and social bias perturbations to the patient agent [7]. *EasyMED* wraps a prompt-driven patient agent in a multi-agent framework with an auxiliary intent-recognition agent and an evaluation agent, and benchmarks the resulting system against human patient actors [18]. In all three cases, the patient itself is only a single behavioral prompt on top of a case description and is thus limited in both realism and steerability. The second family consists of standalone simulator frameworks that expose explicit control over factors such as disclosure or persona. *StateAwarePatient Simulator (SAPS)* models the patient as a controlled state machine with a state tracker, memory bank, and response generator, and introduces inquiry-conditioned disclosure together with explicit metrics for information leakage and passivity [16]. *PatientSim* conditions the simulator on four independent axes: personality, CEFR language proficiency, medical history recall, and cognitive confusion [19]. The resulting personas are evaluated through LLM-as-a-judge scoring and clinician annotation along persona fidelity, NLI (Natural Language Inference)-based factuality, and dialogue-level information coverage and consistency. These frameworks establish that decoupled control and

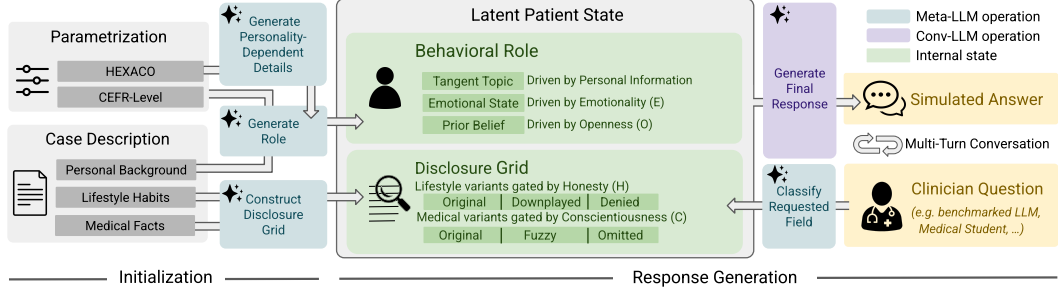


Figure 2: Overview of the *PWP* framework. During **initialization**, the personality parametrization and structured case description are transformed into a latent patient state by a meta-LLM. During **response generation**, the requested case fields are extracted from each incoming question and used to update the disclosure grid. The conversational LLM then generates the patient answer based on the latent role, the currently accessible information, and the dialogue history. The answer is returned to the external system, closing the interaction loop.

explicit disclosure mechanisms are valuable, but none of them ground personality variation in a psychometrically validated theory, and no prior work has compared representative simulators from both families under a shared evaluation protocol.

Modeling personality. Modeling patient personality is central to realistic clinical simulation because patients do not only differ in symptoms, but also in how they communicate, recall, trust, and disclose information during an encounter. Modeling this explicitly is therefore necessary if simulated interactions should expose the same challenges that clinicians and medical assistants face with real patients. The HEXACO model defines personality structure through six dimensions derived from cross-linguistic lexical studies [20]. Its axes capture sincerity and truthfulness versus deceptiveness and manipulation (*Honesty-Humility*), emotional stability versus anxiety and distress reactivity (*Emotionality*), reserve versus talkativeness and social engagement (*eXtraversion*), patience and cooperativeness versus irritability and skepticism (*Agreeableness*), diligence and attention to detail versus carelessness and forgetfulness (*Conscientiousness*), and curiosity and unconventionality versus conventional thinking (*Openness*). Compared with the widely used Big Five personality traits [21], HEXACO preserves rough analogues of *Extraversion*, *Conscientiousness*, and *Openness*, but introduces a distinct *Honesty-Humility* dimension and reconfigures *Agreeableness* and *Emotionality* relative to Big Five *Agreeableness* and *Neuroticism*. For a patient simulator, these distinctions are particularly useful: *Honesty-Humility* provides a principled anchor for personal information disclosure, *Emotionality* isolates distress expression from *Agreeableness*, and the six axes have been empirically linked to health-relevant behavior in a large meta-analysis [22]. Recent work has shown that HEXACO-conditioned prompting can systematically shift LLM generation behavior across models and tasks [23], supporting its use as the personality framework for our simulator. For our work, we drew inspiration from HEXACO dimensions of personality and developed a simulation approach that allows control over behavior expression along similar dimensions. We did not use the original HEXACO survey questions or adjective collections.

3 Simulating Realistic Patients

We introduce *PWP*, a framework for simulating realistic and varied patient dialogue with a configurable behavioral personality and explicit control over the underlying information at each turn. The framework proceeds in two stages, illustrated in Figure 2. During *initialization*, a structured case description and a personality parametrization are transformed into a latent patient state comprising (i) a *behavioral role* that the simulator adheres to throughout the dialogue, and (ii) a *disclosure grid* that controls how facts are revealed across turns. During *response generation*, each incoming question is mapped to the case fields it requests. This both updates the disclosure grid and determines which information is visible when the patient utterance is generated. Information not explicitly requested is excluded from the generation prompt entirely, enforcing non-oversharing by construction. Questions can stem from any external system; in a benchmarking setup, this is the clinician AI under evaluation.

Case description. A simulated patient is grounded in a structured case description $\mathbf{x} = (x_1, \dots, x_m)$, where each entry x_i is a single fact identified by a named field. A field may carry multiple facts (e.g., `chiefcomplaint`: [high blood pressure; palpitations]). The set of fields $\mathcal{F}$ is partitioned into three disjoint semantic subgroups. We differentiate between personal background ($\mathcal{F}_{\text{per}}$), sensitive lifestyle habits ($\mathcal{F}_{\text{life}}$), and clinical facts ($\mathcal{F}_{\text{med}}$). For a subset $S \subseteq \mathcal{F}$, $\mathbf{x}_S$ denotes the corresponding sub-vector of facts. The full list of fields is given in Appendix A.

Personality parametrization. Patient behavior is controlled by a parametrization vector

$$\boldsymbol{\theta} = (h, e, x, a, c, o, l) \in \{1, 2, 3\}^7,$$

where the first six components are the axes of the HEXACO model of personality structure (Honesty-Humility, Emotionality, eXtraversion, Agreeableness, Conscientiousness, Openness), and l is a language-proficiency level mapped to the CEFR bands. Each component takes one of three discrete values (low, medium, high).

The HEXACO axes are mapped to simulator mechanisms based on their behavioral connotation. Honesty-Humility and Conscientiousness are the two axes most directly tied to factual disclosure: low honesty implies concealment of socially undesirable habits, and low conscientiousness implies imprecise recall of clinical detail. These two axes therefore gate the disclosure grid (lifestyle and medical fields, respectively). Emotionality, eXtraversion, and Openness primarily shape *how* patients communicate rather than *what* they disclose. We depict them by injecting case-specific content into the behavioral role (distress state, tangent topics, prior beliefs). Agreeableness only influences style.

LLM components. The framework uses two LLM instances: a meta-LLM Φ_{meta} , which performs structured internal operations (classification, generation of latent variables, generation of fact variants), and a conversational LLM Φ_{conv} , which generates the final patient utterances. Each operation uses a dedicated prompt. We denote prompts as $p_{\bullet}$ and list them in Appendix B. Throughout, we write $\Phi_{\text{meta}}(\cdot; p_{\text{CLASS}})$ for invocation of Φ_{meta} with prompt $p_{\bullet}$ and arbitrary input variables.

Behavioral role construction. The behavioral role is assembled from static and dynamic components. The personal background $\mathbf{x}_{\mathcal{F}_{\text{per}}}$ and CEFR level l are used verbatim. Each HEXACO axis contributes a single-sentence descriptor selected by its level in $\boldsymbol{\theta}$:

$$\mathbf{d}_{\boldsymbol{\theta}} \triangleq (d_H(h), d_E(e), d_X(x), d_A(a), d_C(c), d_O(o)),$$

where each $d_{\bullet}(\cdot)$ is a lookup into a fixed table of three short descriptions per axis (cf. Appendix Table 4). For Openness, Emotionality, and eXtraversion set to the highest level, we additionally generate dynamic, case-specific content that a static descriptor cannot provide. For $x = 3$, we draw a profile-consistent tangent topic s_{tangent} from a predefined, age-conditioned list. For $o = 3$, we generate a plausible prior belief s_{belief} representing a self-diagnosis or lay explanation by prompting Φ_{meta} with p_{0_META} . For $e = 3$, we generate a distress state s_{emotion} that affects communication style without altering clinical facts by prompting Φ_{meta} with personal information $\mathcal{F}_{\text{per}}$ and p_{E_META} .

All of these components are then synthesized into a coherent initial behavioral role description:

$$r = \Phi_{\text{meta}}(\mathbf{x}_{\mathcal{F}_{\text{per}}}, l, \mathbf{d}_{\boldsymbol{\theta}}, s_{\text{belief}}, s_{\text{emotion}}, s_{\text{tangent}}; p_{\text{LATENT_ROLE}}).$$

Disclosure grid. Alongside the behavioral role, initialization sets up a disclosure grid that controls how case information is surfaced to Φ_{conv} . Lifestyle and medical subgroups are gated independently. For each atomic fact x_i in either subgroup, we construct three variants:

- **Lifestyle fields** ($\mathcal{F}_{\text{life}}$): (*original, downplayed, denied*), where downplayed variant is generated by $\Phi_{\text{meta}}(x_i; p_{H_DOWNPLAY})$ and the denial is a fixed phrase per field.
- **Medical fields** ($\mathcal{F}_{\text{med}}$): (*original, fuzzy, omitted*), where the fuzzy variant is generated by $\Phi_{\text{meta}}(x_i; p_{C_FUZZY})$ and the omitted variant masks the fact entirely from the prompt.

Which variant is initially active is determined by the gating personality level, summarized in Table 1.

The asymmetry between lifestyle and medical dynamics is deliberate. Honesty-Humility is a stable trait, so a patient who downplays their alcohol intake on the first request will continue to do so. Conscientiousness, by contrast, governs recall, which is naturally a function of how specifically

Table 1: Overview of disclosure at different levels of h and c . The state of the $\mathcal{F}_{\text{life}}$ fields is fixed at initialization. The fields gated by c advance one step along the disclosure ladder on each request.

Subgroup	Level	State and dynamics
$\mathcal{F}_{\text{life}}$	$h = 1$	<i>original</i>
	$h = 2$	Uniform sample over $\{\textit{original}, \textit{downplayed}\}$
	$h = 3$	Uniform sample over $\{\textit{downplayed}, \textit{denied}\}$
$\mathcal{F}_{\text{med}}$	$c = 1$	<i>original</i> on first targeted request
	$c = 2$	<i>fuzzy</i> $\rightarrow$ <i>original</i> over successive requests
	$c = 3$	<i>omitted</i> $\rightarrow$ <i>fuzzy</i> $\rightarrow$ <i>original</i> over subsequent requests

the patient is prompted. Targeted follow-up questions therefore progressively unmask medical information, recovering first the fuzzy variant and eventually the original. To our knowledge, this dynamic gating is novel in the patient simulation literature and provides a more faithful model of recall under targeted questioning than static-disclosure approaches.

Response generation. At turn t of the simulated multi-turn dialogue, the framework receives a question q_t . First a meta-operation determines which fields are required to answer it:

$$R_t = \Phi_{\text{meta}}(q_t, \mathcal{F}; p_{\text{CLASS}}) \subseteq \mathcal{F}.$$

Only fields in R_t become candidates for inclusion in the generation prompt. Fields not in R_t are excluded entirely, enforcing non-oversharing by construction. For each candidate field, the disclosure grid determines *which variant* is shown. Fields in $\mathcal{F}_{\text{life}} \cap R_t$ retain the variant fixed at initialization. Fields in $\mathcal{F}_{\text{med}} \cap R_t$ are gradually being revealed on each request (cf. Table 1). We denote the final view of the case description at turn t as $\tilde{\mathbf{x}}_t$.

Finally, the conversational model generates the patient utterance, conditioned on the latent role, the currently accessible information, the conversation history, and the incoming question:

$$y_t = \Phi_{\text{conv}}(r, \tilde{\mathbf{x}}_t, H_{t-1}, q_t; p_{\text{SYS}}).$$

The resulting response is grounded in the underlying case description, shaped by the configured behavioral role, and constrained by the current disclosure and recall state.

4 Experimental Setup

We evaluate *PWP* in two complementary studies: The *realism study* (Section 5.1) compares several simulators under their respective neutral default configurations, isolating realism from any personality-specific effect and providing the first medical simulator comparison under a shared evaluation protocol. The *extreme-personality study* (Section 5.2) instantiates the framework at the extreme of each HEXACO axis to test whether configured personality traits are recoverable from interaction traces and whether the resulting conversational behavior is genuinely diverse.

To judge the realism of the simulators we recruited seven resident clinicians with 1-5 years of working experience to evaluate transcripts. Annotation was carried out through a custom labeling interface in which each clinician was presented with 30 transcripts drawn from a stratified pool spanning 10 clinical cases. For the annotation study, we restrict the compared simulators besides PWP and the Human Rephrase to AgentClinic (simple prompt-based) and PatientSim (explicitly parametrizable) to keep per-simulator sample sizes sufficiently high and retain one representative of each prevailing simulator design, divided across a realism judgement task and a personality trait evaluation task. Full details of the annotation study are specified in Appendix G.

Clinician aligned protocol. To enable a fair comparison across simulators, we run all conversations under a shared *clinician-aligned* protocol: the recorded physician utterances are treated as a fixed clinician track, and at every turn the simulator being evaluated produces the patient response in place of the real patient. This removes variance from clinician-side behavior while preserving the natural turn structure of an authentic encounter. Because a simulated patient response can deviate from the recorded one, each subsequent doctor turn is realigned to the preceding simulated utterance through a minimal edit using Gemini-3.1-Flash-Lite (e.g., answering a question of the simulator before proceeding as in the transcript; cf. Prompt B.8).

Dataset. Both realism and extreme-personality experiments use 22 encounters from ACI-Bench [24], a dataset of recorded conversations between physicians and trained patient actors paired with structured patient profiles. The profile instantiates $\mathbf{x}$, while the recorded transcript provides the fixed clinician track used for the clinician-aligned protocol. Exact case IDs are listed in Appendix C.

Model selection. The framework relies on two distinct LLMs. A meta-LLM Φ_{meta} for internal operations, and a conversational-LLM Φ_{conv} for utterance generation. We select the models used in our experiments from a pool of ten open- and closed-weight LLMs spanning multiple sizes and providers by conducting a feasibility study with task-specific probes targeting each role. The LLM candidates, probe definitions, and scores are reported in Appendix D. Based on the results, we use Ministral-3-14B as Φ_{meta} and the more capable Gemini-3.1-Flash-Lite as both Φ_{conv} and as the LLM judge in the following.

Meta-prompt tuning. The meta-prompt that synthesizes the latent role r_0 is a central control point of the framework and has a large effect on downstream behavior [25–27]. To account for this, we optimize $p_{\text{LATENT_ROLE}}$ against a penalty that jointly measures information control and personality control across formulations of the resulting latent role description. The resulting compact prompt matches longer evolved candidates at a fraction of the input-token cost (cf. Appendix E).

5 Results

5.1 Realism Study

We compare seven patient simulators in a neutral default configuration (cf. Appendix F). For this we contrast the simulators against two controls. *Human Actor* denotes the original answers from the recorded encounter between physician and trained patient actor, while *Human Rephrase* rephrases those same recorded answers with Gemini-3.1-Flash-Lite using Prompt B.9. The Human Rephrase was added to evaluate the effects of how a naive LLM generated text is perceived, even if the basis for that text is simply a rephrasing of a real human utterance. The remaining sources are implementations released with their respective papers: *AgentClinic* [7], *PatientSim* [19], *StateAwarePatient* [16], *CraftMD* [8], *VirtualPatient* [18], and our own *PatientsWithPersonality*.

Clinician judgments of realism. Figure 3 reports the fraction of transcripts classified as *real* and, for transcripts flagged as simulated, the reasons annotators gave (if any). PWP is classified as real in 48.9% of the cases, directly behind the human actor (52.5%) and ahead of both the rephrased actor (37.5%) and all other simulators ($< 30\%$). PWP was judged as realistic 11%p more often than a naive LLM rephrasing of the original human response, showcasing how successful our framework is in removing the default LLM style of generated outputs. Crucially, compared to the other simulators, PWP was flagged as “too informative” only about half as often (20.8% vs 34.4% and 40.0%). Human Actor and Human Rephrase transcripts are flagged disproportionately often as *inconsistent with prior context*, which can reflect the inherent variability of unprompted patient speech. We include further study statistics in Appendix J.

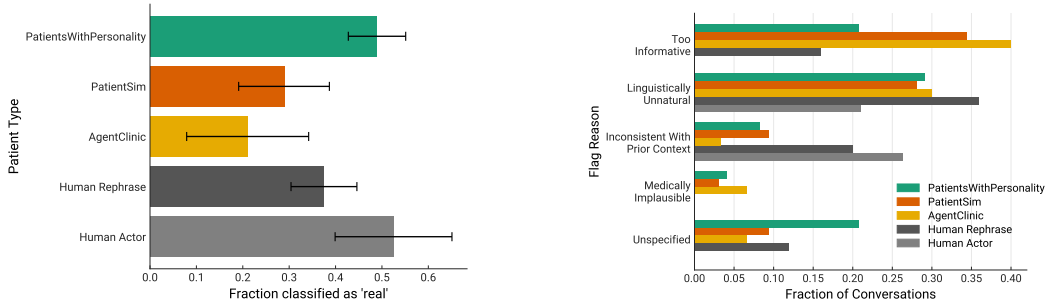


Figure 3: **Left:** Clinicians judge PatientsWithPersonality conversations as real at similar rates as the original recorded encounter and well above the rates of other simulators. **Right:** Other simulators are flagged as “too informative” about twice as often as PatientsWithPersonality.

Finding: Clinicians classify PWP transcripts as real nearly as often as actual recorded encounters, clearly ahead of every competing simulator.

Quantitative dialogue metrics. The clinician-aligned protocol pairs every simulated patient turn with its human actor counterpart. Treating the recorded human track as the reference, we report the per-conversation signed difference (sim minus real) on four dialogue metrics in Figure 4. Values near zero indicate agreement with the recorded encounter. The metrics are the *token count*, capturing the verbosity of the simulator; *lexical diversity*, where lower values indicate more repetitive vocabulary; *domain term count*, capturing how much clinical terminology the patient uses; and *readability* (Flesch–Kincaid grade level [28]), where higher values indicate more complex speech. PWP and PatientSim both stay close to the reference across all four metrics, with PWP clearly closest to the human actor on lexical diversity. The remaining simulators deviate further, producing utterances that are more lexically uniform, contain more clinical jargon, and are more difficult to read than the human actors and PWP.

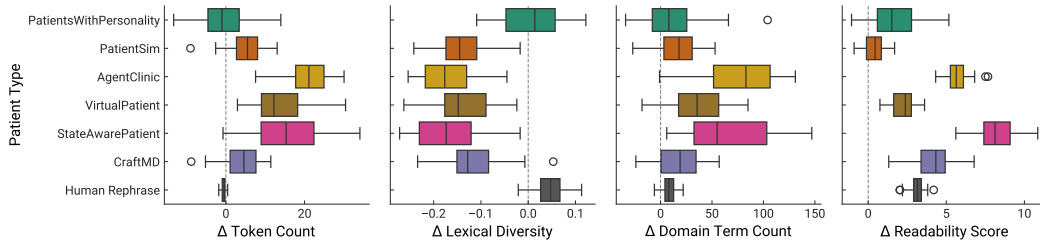


Figure 4: Per-conversation differences (sim – real) from the recorded patient track. Values closer to zero indicate closer agreement with the recorded transcripts. PatientsWithPersonality is closest to the original recorded encounter across most metrics and especially on lexical diversity and token count, driving its realism scores.

Finding: PWP tracks the recorded encounters most closely, with significantly tighter agreement on lexical diversity than any other simulator.

Disclosure behavior. To measure information sharing behavior, we mark a case field as *disclosed* at the first turn at which the simulator’s utterance contains content matching that field. We then calculate the cumulative fraction of fields disclosed up to turn t , averaged over all conversations (cf. Figure 5).

Human Actor, Human Rephrase, and PWP all have similar disclosure behaviors showing both how realistic our information control is and that the human rephrase does not destroy or invent new information. The remaining simulators saturate both earlier and at higher levels, indicating they both more quickly share information and provide information that was never provided in the original conversations. We provide additional details on the analysis and more fine-grained metrics in Appendix H.

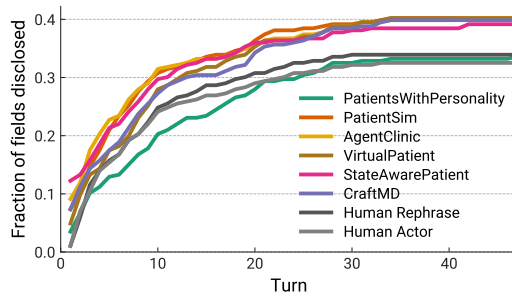


Figure 5: Cumulative fraction of case fields disclosed, averaged over conversations. PatientsWithPersonality discloses at the same rate as both the original Human Actor and the Human Rephrase.

Finding: PWP exhibits the most realistic information-disclosure behavior, reflecting our design goal of on-demand rather than volunteered disclosure.

5.2 Extreme-Personality Study

To isolate the effect of each personality axis, we instantiate six PWP configurations in which exactly one HEXACO axis is set to its extreme value (3) while the remaining five are kept at a low level (1), with the language level fixed to intermediate. We pair these with six PatientSim configurations chosen to expose the closest available behavioral analogue. The complete configuration list is given in Appendix I.

Table 2: Absolute differences to real parametrizations on the HEXACO axes recovered by human raters and the autorater. Best (i.e. lowest) mean values are bolded.

Patient Simulator	Rater	H	E	X	A	C	O	Mean
PWP	Autorater	0.333	0.167	1.000	0.367	0.300	0.067	0.372
	Human	0.468	0.308	0.756	0.427	0.680	0.391	0.505
	Mean	0.401	0.238	0.878	0.397	0.490	0.229	0.439
PatientSim	Autorater	0.200	1.167	0.967	0.667	0.433	0.633	0.678
	Human	0.638	1.208	0.533	0.885	0.820	0.870	0.826
	Mean	0.419	1.188	0.750	0.776	0.627	0.752	0.752
PatientSim – PWP	Mean	0.018	0.950	−0.128	0.379	0.137	0.523	0.313

Diversity of conversational behavior. First, we tested if the configured HEXACO traits are recoverable from interaction traces. We tasked both our clinicians and an LLM autorater to estimate the personality level for each HEXACO trait (i.e. 1, 2, 3) from both PWP and PatientSim. As seen in Table 2, both clinicians and the autorater predicted PWP personality levels more accurately than PatientSim, with average absolute differences of 0.505 vs. 0.826 (clinicians) and 0.372 vs. 0.678 (autorater). Averaged across raters, PWP achieves a mean absolute difference of 0.439 compared to 0.752 for PatientSim. Looking at individual traits, the gap is most pronounced on Emotionality (0.950) and Openness (0.523), where PatientSim struggles to faithfully render the configured levels, while Honesty-Humility is recovered comparably well by both simulators (difference of 0.018). Extraversion is the only axis on which PatientSim outperforms PWP (−0.128), though both simulators show large absolute errors on this trait suggesting it is difficult to estimate. The autorater and human means also agreed closely across both simulator families and all six axes ($r = 0.87$, ICC(2, 1) = 0.85; Figure 6). Complete study details are in Appendix G.

To then characterize the behavioral footprint of each configuration, we embed every patient utterance with a sentence encoder and project the embeddings into a shared 2D PCA space estimated jointly across both simulators (Figure 7). The *mean pairwise distance* between projected utterances is higher for PWP than for PatientSim (14.12 ± 7.01 vs. 11.78 ± 6.62), indicating greater average dispersion. The *convex hull area* of the projected points is likewise substantially larger (888.50 vs. 577.06), reflecting a wider extent of the dialogue space reached. Finally, the *Vendi score* [29], a similarity-based effective number of modes, is higher for PWP (5.48 vs. 4.23), indicating that its outputs cluster around more distinct conversational patterns rather than collapsing into a few. Taken together, these metrics show that our framework is able to generate more diverse and varied sentences while still maintaining a high degree of control.

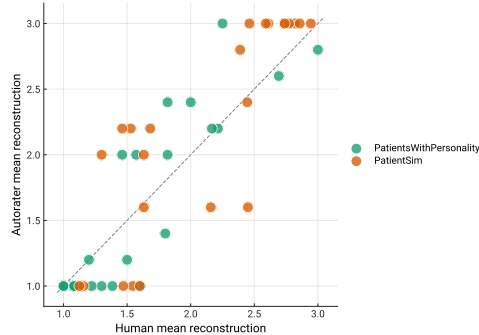


Figure 6: Per-axis HEXACO reconstruction by the autorater versus clinician annotators on the personality task transcripts. Human and autorater agreed closely across both frameworks achieving $r = 0.87$. Dashed line is the identity line.

Finding: PWP exhibits a highly diverse conversational behavior across personas.

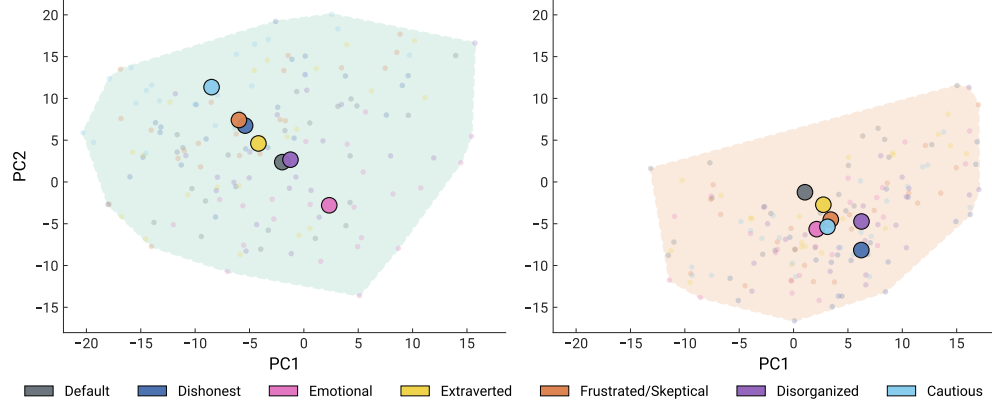


Figure 7: Sentence-embedding PCA projection of patient utterances. **Left:** PatientsWithPersonality. **Right:** PatientSim. Small dots are individual utterances, large markers are per-persona means. PatientsWithPersonality encapsulates a markedly larger space than PatientSim reflecting the increased behavioral diversity that can be simulated with our framework.

Selective disclosure. We also test two information-handling configurations: $h = 3$ (*Dishonest*) and $c = 3$ (*Disorganized*), which gate $\mathcal{F}_{\text{life}}$ and $\mathcal{F}_{\text{med}}$ respectively. Each axis should suppress recall on its own subgroup while leaving the other untouched. We measure per-conversation *fact recall* (fraction of facts appearing in patient utterances) separately for $\mathcal{F}_{\text{life}}$ and $\mathcal{F}_{\text{med}}$. As expected, *Dishonest* suppresses $\mathcal{F}_{\text{life}}$ recall (-0.133) without affecting $\mathcal{F}_{\text{med}}$ (0.011), while *Disorganized* drops $\mathcal{F}_{\text{med}}$ (-0.152) while preserving $\mathcal{F}_{\text{life}}$ (-0.015).

Finding: Our disclosure framework effectively and selectively controls information.

6 Discussion

We introduced PWP to address the two failure modes that have limited the realism of LLM-based patient simulators: *oversharing* of information the doctor never asked for, and *uniformity* across cooperative, knowledgeable, and ultimately indistinguishable patients. PWP resolves both by design. The query-conditioned disclosure grid removes unrequested fields from the prompt altogether, enforcing on-demand disclosure by construction rather than by instruction. The HEXACO-parametrized latent role couples personality variation to interpretable, recoverable conversational behavior across distinct configurations within a single framework.

Our shared conversation protocol allows us to conduct the first cross-simulator comparison in the medical LLM literature and establish that our design improves upon other simulators across every dimension that matters for downstream evaluation. Clinicians judge PWP conversations as real at rates comparable to the actual recorded encounters and clearly above all competing simulators. Furthermore, among the seven simulators evaluated, PWP exhibits the most controlled information-disclosure dynamics, again closely matching the rates of the original encounter thus addressing the major failure mode of all previous simulators. Finally, configured HEXACO traits are reliably recoverable from interaction traces and propagate into a conversational footprint that is substantially more diverse than the only other parametrizable baseline along every measure.

Limitations. The encounters underlying our evaluation were recorded with trained patient actors rather than with real patients in genuine clinical visits. While this allowed for a diverse and controlled set of cases with high quality dialogues, a residual distance from authentic clinical speech remains. Realism judgments were further provided by clinicians whose primary working language is German, which may introduce subtle scoring differences relative to native-speaker raters, although the close agreement with the LLM autorater suggests any such effect is modest. Importantly, realistic simulations should complement rather than replace evaluations with human participants, and con-

sequential deployment decisions should continue to rest on studies involving real patients. PWP is best positioned as a screening layer upstream of such studies, narrowing the space of models and configurations that warrant the cost and ethical scrutiny of human evaluation.

Conclusion PatientsWithPersonality is a controllable patient simulation framework that improves the realism, diversity, and information-disclosure behavior of LLM-based virtual patients. By combining query-conditioned disclosure with HEXACO-grounded personality parametrization, PWP mitigates oversharing while producing behaviorally distinct personas whose configured traits are recoverable from interaction traces. These results suggest that realistic and steerable patient simulators can provide a strong foundation for scalable, clinically meaningful evaluation of medical LLMs before costly human evaluation studies.

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A Structured Case Description Fields

Table 3 enumerates the named fields of the structured case description $\mathbf{x}$ used by *PWP*, together with their assignment to the three semantic subgroups $\mathcal{F}_{\text{per}}$, $\mathcal{F}_{\text{life}}$, and $\mathcal{F}_{\text{med}}$. Personal fields are always visible to the conversational model. Lifestyle fields are subject to an honesty-humility-dependent disclosure mask with three variants (original, downplayed, denied). Medical fields are subject to a conscientiousness-dependent recall mask (original, fuzzy, omitted) that is relaxed when the corresponding content is explicitly elicited by the doctor.

Table 3: Named fields of the structured case description and their assignment to the subgroups $\mathcal{F}_{\text{per}}$ (personal background), $\mathcal{F}_{\text{life}}$ (sensitive lifestyle habits), and $\mathcal{F}_{\text{med}}$ (clinical facts). Field names match the keys used in the reference implementation.

Field	Content
<i>$\mathcal{F}_{\text{per}}$ — Personal background (always visible)</i>	
age	Patient age in years.
gender	Self-identified gender.
marital_status	Relationship status (e.g., single, married, widowed).
children	Number and/or presence of children.
living_situation	Housing and household composition.
occupation	Current or most recent occupation.
insurance	Insurance status or provider type.
arrival_transport	Mode of arrival at the encounter (e.g., walk-in, ambulance).
chiefcomplaint	Primary complaint stated on presentation.
<i>$\mathcal{F}_{\text{life}}$ — Sensitive lifestyle habits (honesty-humility-gated disclosure)</i>	
tobacco	Tobacco use and frequency.
alcohol	Alcohol consumption pattern.
illicit_drug	Use of non-prescribed or illicit drugs.
sexual_history	Relevant sexual history.
exercise	Physical activity routine.
<i>$\mathcal{F}_{\text{med}}$ — Clinical facts (conscientiousness-gated recall)</i>	
allergies	Known allergies and associated reactions.
family_medical_history	Relevant medical history in first-degree relatives.
medical_device	Implants or assistive medical devices.
medical_history	Past diagnoses, conditions, and procedures.
present_illness_positive	Affirmed symptoms of the current illness.
present_illness_negative	Explicitly denied symptoms (pertinent negatives).
pain	Pain location, character, and severity.
medication	Current medications and dosages.

B Prompt Templates

This section reproduces the prompt templates used by *PWP* at each stage of the simulation loop introduced in Section 3 and throughout the evaluation. Placeholders substituted at call time are shown in <purple>; all other content is rendered verbatim.

Prompt B.1: Latent role construction (LATENT_ROLE)

You are building an internal latent role card for a virtual patient. Use the data exactly as provided and do not invent medical facts. The role should be practical, grounded, and focused on communication tendencies under clinical questioning.
Return only <role>...</role>.

Personal Information: <personal_information>

HEXACO Personality: <hexaco_personality>

Prompt B.2: Conversational system prompt (SYS)

<latent_role>

Patient Profile: <structured_information>

Provide your answer to the doctor's inquiry between the tags <response> and </response>. Do not include any other text in your answer.

Prompt B.3: Emotional-state generator (E_META)

Generate an emotional state of distress for this personal profile: <personal_information>

Requirements:

- The state must directly affect communication style, not factual content.
- Specify how the distress alters speech (e.g., avoidance, repetition, irritability, catastrophizing).
- Keep the description short (1–2 sentences).

Output the emotional state description between the tags <state> and </state>.

Prompt B.4: Prior-belief generator (O_META)

Based on the complaint "<chiefcomplaint>", generate a plausible but specific self-diagnosis that a non-medical person might find on an internet forum or from a 'friend.' This diagnosis should be something the patient can fixate on dogmatically. Return the self-diagnosis between the tags <prior_belief> and </prior_belief>.

Prompt B.5: Lifestyle downplay generator (H_DOWNPLAY)

You are a linguistic assistant that converts medical data into casual, minimized euphemisms. Follow the format of the examples below exactly. Provide only the tag and the content.

Input: 'drinks 6 beers a day'

Output: <phrase>drinks a few beers</phrase>

Input: 'Marijuana, smoked, every weekend'

Output: <phrase>uses a little weed sometimes</phrase>

Input: '<leisure_info>'

Output:

Prompt B.6: Medical fuzzifier (C_FUZZY)

You are a linguistic assistant that converts medical data into a fuzzy, broader description. Transform the actual content while keeping the same general meaning. Provide only the tag and the content.

Input: family_medical_history: 'Mother had breast cancer at age 45'

Output: <phrase>mother had cancer, not sure what kind or when exactly</phrase>

Input: pain: '7–8'

Output: <phrase>pretty strong pain</phrase>

Input: <medical_info>

Output:

Prompt B.7: Relevant-field classifier (CLASS)

You are analyzing a medical consultation conversation to identify which fields from the patient’s case description are required to answer the doctor’s question.

Available case description fields: `<available_fields>`

Doctor’s Question: `<question>`

Task: Identify case description fields that contain information asked about in the doctor’s question. Return only fields that are explicitly mentioned or clearly implied by the question. Return a list of relevant field names from the available fields. Return an empty list if the question is too vague or no fields are directly relevant.

Prompt B.8: Doctor-utterance alignment (PWP _DOC_ALIGN)

Patient message: `<patient_last_message>`

Original doctor response: `<response>`

Task: Edit the doctor response so it is consistent with the patient message.

Constraints:

- Make the smallest possible changes (minimal edits).
- Keep wording, structure, and length as close as possible to the original.
- Only modify parts that are inconsistent or fail to address the patient message.
- If the patient asks a question, answer it briefly within the existing response.
- Do not add new information or expand the response.

Return only the revised doctor response.

Prompt B.9: Human rephrase (PWP _REPHRASE)

System prompt:

You are a helpful assistant that rephrases patient responses. Maintain the same meaning, information content, and tone.

User prompt:

Rephrase the following patient response. Keep the response length similar to the original response:

`<real_response>`

Provide your answer to the doctor’s inquiry between the tags `<response>` and `</response>`. Do not include any other text in your answer.

C Datasets

ACI-Bench. ACI-Bench [24] provides full doctor–patient conversations paired with the corresponding clinical notes. We restrict ourselves to cases for which both a structured patient profile and a clean transcript with strictly alternating patient and doctor utterances are available, yielding the 22 IDs VS000, VS007, VS013, VS018, VS024, VS037, VS001, VS008, VS014, VS019, VS030, VS038, VS002, VS009, VS015, VS022, VS035, VS005, VS012, VS017, VS023, VS036. The structured profile instantiates the case description `x`, while the transcript provides the fixed clinician track used in the doctor-aligned protocol. We use ACI-Bench for the *realism study* (cf. Section 5.1) and the *extreme-personality study* (cf. Section 5.2).

D LLM Feasibility Study

The feasibility study identifies which LLMs can reliably perform the operations required by the meta- and conversational model roles in PWP. Candidates span proprietary frontier models, which are only accessible via remote API, and smaller open-weight models, which can be served locally with batched inference and are therefore substantially cheaper and faster to invoke for the meta-operations that run on every turn. All probes are executed on a single fixed case description with five replications

Table 4: HEXACO level descriptors assembled into `<hexaco_personality>` for the latent-role construction prompt. The state variable `<state>` is generated by prompt E_META, `<prior_belief>` is generated by prompt O_META.

Axis	Level	Descriptor
Honesty-Humility (H)	1	You are transparent and honest, sharing all habits including substance use or non-compliance openly.
	2	You are generally truthful but hesitant to share embarrassing habits and tend to downplay them.
	3	You are actively hiding specific habits and may provide false information about your medical history.
Emotionality (E)	1	You are emotionally detached and calm. Report facts objectively without expressing fear or distress.
	2	You are concerned about your health.
	3	<code><state></code>
eXtraversion (X)	1	You are passive and reserved. Only answer exactly what is asked using as few words as possible.
	2	You have a natural conversational flow. Provide concise but complete answers that stay strictly on topic.
	3	You are talkative and inquisitive. Include personal anecdotes and ask the doctor questions about the process.
Agreeableness (A)	1	You trust the medical staff and the process. Be helpful and cooperative.
	2	You are guarded and suspicious. Do not volunteer information unless the doctor asks a very specific question.
	3	You are frustrated and skeptical of the doctor. You are easily annoyed and may bring up past bad experiences.
Conscientiousness (C)	1	You remember dates, times, and dosages for all medical events.
	2	Your memory for dates and dosages is a bit fuzzy.
	3	You struggle to remember when symptoms started or what medications you take.
Openness (O)	1	You appreciate scientific explanations and are open to any logical medical advice provided.
	2	You are hesitant about new technology. You prefer standard, tried-and-true treatments or home remedies.
	3	You are convinced you already know what is wrong. Based on your own research, you assume that you have: <code><prior_belief></code> . Dismiss other ideas.

per candidate; reported numbers are per-model averages across replications and, where applicable, across probe sub-items.

Meta-model selection. The meta-model Φ_{meta} is probed on three operations it must perform reliably: generating the lifestyle downplay and denial masks that populate G_i^{life} ; sampling the context-dependent distress state s_{emotion} used for high-emotionality personas; and returning, for each incoming doctor question, the relevant-field subset R_t . The last operation dominates cost — it is invoked on every doctor turn and must emit well-formed structured output at low latency — so only candidates that can be served locally with batching are considered. Per-probe scores are reported in Table 5. Minstral-3-14B achieves the highest average score and is selected as Φ_{meta} .

Table 5: Meta-LLM probe scores on the fixed reference case, averaged over five replications. Bold marks per-column maxima. All candidates are open-weight and hosted locally with batched vLLM serving.

Model	Open	Downplay	Emotional State	Field Classification	Avg. Score
Minstral 14B	✓	0.867	1.000	0.500	0.789
GPT-OSS 20B	✓	0.733	1.000	0.428	0.720
Minstral 8B	✓	0.611	1.000	0.500	0.704
Qwen 3 8B	✓	0.667	0.833	0.463	0.654

Conversational-model selection. The conversational model Φ_{conv} is probed on two tasks: producing a single-turn patient utterance grounded in $\mathbf{x}$ under a default persona, as a smoke test for

conversational viability; and generating utterances under contrasting HEXACO configurations whose trait differences must be recoverable by an independent judge. All candidates saturate the basic-simulation probe, so the HEXACO-steering probe is the discriminating signal. Per-probe scores are reported in Table 6. Claude Opus 4.6 attains the highest average score but at markedly higher per-token cost and per-call latency than the rest of the pool. Gemini-3.1-Flash-Lite is the second-ranked candidate while offering substantially lower API cost and faster inference, and is selected as Φ_{conv} . The same model and configuration is also used for all LLM-judge calls in the downstream evaluation, so that the judge and conversational roles remain consistent.

Table 6: Conversational-LLM probe scores on the fixed reference case, averaged over five replications. Bold marks per-column maxima. **Open** = open-weight model that can be hosted locally; remaining candidates are proprietary and API-only. Pricing reflects weighted averages from OpenRouter³, given as input/output USD per 1M tokens.

Model	Open	Basic Simulation	Personality Steering	Avg. Score	In/Out (\$/1M tok)
Claude Opus 4.6	–	1.000	0.733	0.867	2.03 / 25.00
Gemini 3.1 Flash Lite	–	1.000	0.633	0.817	0.180 / 1.50
GPT-5.2	–	1.000	0.600	0.800	1.05 / 14.00
MiniMax M2.5	✓	1.000	0.567	0.783	0.097 / 1.23
DeepSeek V3.2	✓	1.000	0.533	0.767	0.202 / 0.469
Ministral 14B	✓	1.000	0.520	0.760	0.190 / 0.207
Ministral 8B	✓	1.000	0.467	0.733	0.162 / 0.181
Qwen 3 8B	✓	1.000	0.433	0.717	0.094 / 0.445
Gemini 3 Flash	–	1.000	0.400	0.700	0.284 / 2.99
GPT-OSS 20B	✓	1.000	0.280	0.640	0.039 / 0.176

E Meta-Prompt Tuning

The latent-role construction prompt is the main persona-relevant conditioning signal that Φ_{conv} receives on every subsequent turn. Therefore phrasing-level variation in its output compounds over the dialogue and can suppress or mischaracterize individual HEXACO axes, distort personal-information exposure, or inject medical content absent from $\mathbf{x}$. Hand-tuning this prompt does not scale across the 3^7 different configurations, so we instead optimize it end-to-end against a small set of doctor probes applied to patient profiles extracted from role-played patient-physician interviews Fareez et al. [30].

Dataset. Ten distinct patient profiles $\{\mathbf{x}^{(k)}\}_{k=1}^{10}$ are drawn from the interview corpus. For each profile, personality configurations $\theta^{(k,j)} = (h, e, x, a, c, o, l)^{(k,j)} \in \{1, 2, 3\}^7$ are sampled with uniform probability across all levels. The resulting tuning corpus

$$\mathcal{D} = \{(\mathbf{x}^{(k)}, \theta^{(k,j)})\}_{k=1, \dots, 10; j=1, \dots, n_k}, \quad \sum_k n_k = 161,$$

is split by *profile*: profiles $k \in \{1, \dots, 7\}$ form the training pool $\mathcal{D}_{\text{train}}$ (111 rows) and profiles $k \in \{8, 9, 10\}$ form the held-out evaluation pool $\mathcal{D}_{\text{eval}}$ (50 rows). Blocking by profile prevents profile-specific phrasing from leaking across the split.

Optimizer. We tune the prompt with CAPO [31], an evolutionary prompt optimizer that maintains a population of candidate prompts, alternates LLM-driven mutation and crossover steps, and eliminates candidates early via racing. We run CAPO for 99 steps with sequential-block evaluation, 20 subsamples per step and no few-shot examples. The meta-LLM (Ministral-3-14B) is served via vLLM (temperature 0.05, 1024-token output budget). A reward cache deduplicates evaluations of repeated (prompt, row) pairs across optimizer steps.

Reward. For each candidate prompt π and each row $(\mathbf{x}, \theta) \in \mathcal{D}$, we instantiate a PWP whose latent role r_0 is generated by π and run it against four probe questions designed to expose distinct trait combinations: (i) *symptom history and self-management* (Extraversion, Emotionality); (ii) *lay causal explanation* (Openness, Emotionality); (iii) *worry elicitation* (Agreeableness, Extraversion);

³Accessed 05/06/2026.

(iv) *reassurance challenge* (Agreeableness, Openness). Let $P_{\text{info}}(\pi, \mathbf{x}, \theta)$ be the fraction of extracted facts judged *Hallucination* against $\mathbf{x}$; $P_{\text{pers}}(\pi, \mathbf{x}, \theta)$ the mean absolute distance between the configured HEXACO levels and the levels reconstructed by the LLM judge from the probe responses, restricted to the axes targeted by each probe; and $P_{\text{pi}}(\pi, \mathbf{x}, \theta)$ the share of personal-information fields judged *Missing* from the generated latent role. The reward is an equally weighted sum of these three penalties:

$$R(\pi, \mathbf{x}, \theta) = -[P_{\text{info}}(\pi, \mathbf{x}, \theta) + P_{\text{pers}}(\pi, \mathbf{x}, \theta) + P_{\text{pi}}(\pi, \mathbf{x}, \theta)]$$

As a conversational LLM and LLM-judge we use Gemini-3.1-Flash-Lite, the same judge model used in the downstream evaluation, so the optimization target is consistent with the main evaluation protocol.

Results. Figure 8 shows the per-step score of every prompt surviving into the final population: a handful of early candidates are pruned within a few steps, after which the population converges around a score of -1.1 , and CAPO mutations drift toward substantially longer prompts. Figure 9 decomposes the held-out reward into its two dominant penalty components. Prompt 318 reduces the mean information penalty well below the short prompts, but this gain does not carry over to the personality-penalty axis. On $\mathcal{D}_{\text{eval}}$ the compact prompt 2 nevertheless attains the highest mean reward of the final population, narrowly ahead of prompt 318. The two prompts are shown explicitly in Table 7: prompt 318 expands into an explicit ED-context specification with nested sub-requirements, whereas prompt 2 states the task in two short sentences. Given the length difference and the corresponding savings in meta-model input tokens spent on every simulated patient, prompt 2 is selected as the final meta-prompt.

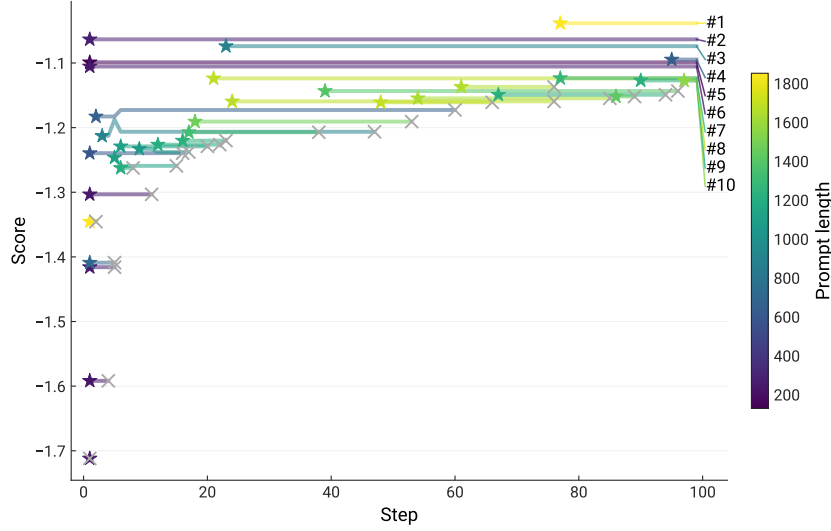


Figure 8: Reward of every prompt at the end of each step. The trajectory is colored by prompt length; stars mark the step at which a prompt entered the population and grey crosses mark elimination.

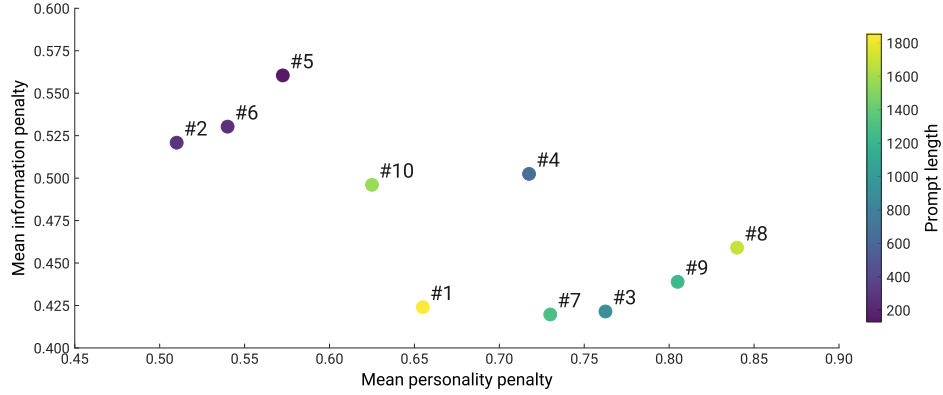


Figure 9: Mean information penalty versus mean personality penalty on $\mathcal{D}_{\text{eval}}$ for every prompt in the final population. Points are labeled by prompt ID and colored by prompt length.

Table 7: The two highest-scoring prompts of the final population after tuning with CAPO.

ID	Prompt text	Avg. R on $\mathcal{D}_{\text{eval}}$
2	You are building an internal latent role card for a virtual patient. Use the data exactly as provided and do not invent medical facts. The role should be practical, grounded, and focused on communication tendencies under clinical questioning. Return only <code><role>...</role></code>.	−1.11
318	Design a meticulously crafted, behaviorally authentic latent role for an advanced emergency department (ED) clinical interview simulator. The role must be constructed from the provided personal history and HEXACO personality framework (Honesty-Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, Openness to Experience), ensuring that all verbal and nonverbal cues—such as speech patterns, emotional cues, and interactional nuances—align seamlessly with real-world patient-clinician dynamics. Key Specifications: Patient Interaction Profile for ED Context: - Define the patient’s communication style (e.g., succinct, meandering, or overly detailed) and confidence level (e.g., uncertain, assertive, or evasive). - Establish tendencies in questioning behavior (e.g., inquisitive, evasive, or compliant). - Determine receptiveness to reassurance (e.g., responsive, indifferent, or conflicted). - Strictly adhere to all factual parameters provided in the patient’s background. Psychological and Behavioral Realism: - Base all behavioral responses solely on the given data, avoiding any speculative clinical interpretations or diagnostic assumptions. - Emphasize observable communication traits during clinical questioning, including: - Likely information-sharing strategies (e.g., open disclosure vs. guarded responses). - Adaptive reactions to clinician tone, pacing, or questioning style (e.g., defensive, cooperative, or disengaged). - Ensure the role reflects how the patient would <i>realistically</i> behave in an ED scenario, grounded in empirical accuracy. Output Structure: Provide the role description exclusively within <code><role></code> and <code></role></code> tags, ensuring strict adherence to the provided constraints. No extraneous commentary or speculative elements are permitted.	−1.14

F Realism Study Configuration

The realism study runs on the 22 ACI-Bench cases. Table 8 enumerates the simulator instances run on each case under the doctor-aligned protocol. The conversational model is Gemini-3.1-Flash-Lite (temperature 1.0, top- p 0.95, low thinking budget) and the meta-model is Ministral-3-14B (temperature 0.05, batch size 16). The same Gemini-3.1-Flash-Lite configuration is used as the LLM judge for all downstream evaluation calls.

Table 8: Simulator instances and parameters used in the realism comparison study. PWP parameters use the seven-tuple (h, e, x, a, c, o, l) defined in Section 3. Identifiers between dashes correspond to the run names emitted by the reference implementation.

Simulator	Parameters	Notes
Human Rephrase	—	Lower bound; paraphrases ground-truth turns.
CraftMD	default	1-sentence layperson responses.
VirtualPatient	default	EasyMED-style prompt with rolling history.
StateAwarePatient	default	8-state controller, three-tier memory.
AgentClinic	no bias	Cognitive/social bias disabled.
PWP	(2, 1, 1, 2, 2, 1, B)	Default neutral persona.
PatientSim	B / plain / recall=high / dazed=normal	Default reference.

G Clinician Study Details

G.1 Annotation Protocol

To assess the realism of our patient simulator and to validate the LLM-as-a-judge, a subset of the evaluation is replicated with human expert annotators (seven resident clinicians) through a custom labeling interface. For this, clinicians were asked to assess both the realism of a conversation and the patient’s personality in two independent tasks.

Realism task. Each clinician was assigned a subset of 10 distinct clinical cases, with each case presented in three conversation variants sampled randomly from five different sources, yielding 30 conversations observed in total. The sources included two authentic conversations (the original dialogue from the ACI dataset (*Human Actor*) and a LLM-rephrasing of it (*Human Rephrase*)) and three simulated conversations (by *AgentClinic* (no bias), *PatientSim* (CEFR B, plain, normal dazedness, high recall), and *PWP* (2, 1, 1, 2, 2, 1, B)). The physicians were informed that for each set of variants, between zero and two versions could be real, with the remainder being simulated. For every conversation variant, annotators provided (i) a binary classification indicating whether they believed the conversation was real or simulated along with a confidence score for this classification on a Likert scale from 1 (guessing) to 5 (certain), (ii) a rating of perceived content realism split into symptom description realism on a Likert scale from 1 (implausible/textbook) to 5 (clinically plausible & natural) and information control on a Likert scale from 1 (volunteers clinical details) to 5 (realistic lay knowledge) and (iii) perceived style realism, each on a Likert scale from 1 (artificial phrasing) to 5 (sounds like a real person). In addition, clinicians were able to flag specific patient utterances within each conversation that they considered unrealistic and to indicate the underlying reason, including unnatural language, inconsistency with prior context, too high informativeness, medical implausibility, or other justifications. Variants were randomly sampled across the five sources and the assignment of conversations to human annotators was structured to include an overlap between annotator pairs (Jaccard Similarity: 0.41).

Personality task. In a second part of the study, clinicians evaluated patient personality traits based on the HEXACO personality model, assigning scores on a scale from 1 to 3. The evaluation set comprised twelve patient configurations: six extreme configurations from PWP and six matched configurations from *PatientSim*, with 6 patients sampled per case without replacement. This extreme-personality study was conducted on the ACI-Benchmark cases (see Appendix I) and was designed to isolate individual HEXACO axes by varying a single component of θ at a time while holding all other axes at their lowest level and fixing language proficiency to CEFR level B. Notably, to avoid bias clinicians were not informed about the underlying parameterization of the personality

configurations and, in particular, were unaware that each case varied only a single axis while the others were held constant. Only *PatientSim* was included for comparison as it is the only other tested simulator that allows explicit variation of patient personality traits. To enable a fair comparison between the two patient simulators, each configuration of PWP was paired with the closest behavioral analogue available within the parameter space of *PatientSim*, as detailed in Table 10. These pairings account for structural differences between the simulators, notably that *PatientSim* does not disentangle Honesty-Humility from recall and does not expose Conscientiousness or Openness as distinct axes. Accordingly, configurations were selected based on the closest overlap in dominant behavioral traits with the targeted HEXACO axis.

H Information Disclosure Analysis

Table 9 reports the disclosure dynamics in more detail. Median time-to-first-disclosure (TTF) and disclosure AUC characterize the pacing of information release. Prompted-disclosure precision measures whether information appears when the doctor explicitly asks for it. We reuse the relevant-field classifier $\Phi_{\text{meta}}(\cdot; p_{\text{CLASS}})$ to identify the requested fields R_t at each turn, and check which of those fields surface in the patient’s answer. Unprompted leakage is the complementary failure mode, measuring the fraction of fields revealed in turns where they were not requested.

Table 9: Disclosure metrics by simulator (mean $\pm$ SEM). Arrows in the column headers indicate the direction of more controlled, on-demand disclosure. Best values (within one SEM) are marked bold.

Simulator	Median TTF $\uparrow$	Disc. AUC $\downarrow$	Prompted-Disc. Precision $\uparrow$	Unprompt. Leak. $\downarrow$
Human Rephrase	6.75 $\pm$ 0.91	0.26 $\pm$ 0.02	0.27 $\pm$ 0.03	0.22 $\pm$ 0.04
PatientSim	6.05 $\pm$ 0.90	0.31 $\pm$ 0.02	0.43 $\pm$ 0.03	0.25 $\pm$ 0.03
AgentClinic	5.93 $\pm$ 0.98	0.31 $\pm$ 0.02	0.41 $\pm$ 0.03	0.20 $\pm$ 0.02
VirtualPatient	7.82 $\pm$ 1.06	0.29 $\pm$ 0.02	0.42 $\pm$ 0.04	0.19 $\pm$ 0.04
StateAwarePatient	6.23 $\pm$ 1.30	0.30 $\pm$ 0.02	0.37 $\pm$ 0.03	0.33 $\pm$ 0.05
CraftMD	8.64 $\pm$ 1.38	0.29 $\pm$ 0.02	0.43 $\pm$ 0.03	0.29 $\pm$ 0.04
PWP (<i>Ours</i>)	9.68 $\pm$ 1.44	0.23 $\pm$ 0.02	0.41 $\pm$ 0.03	0.13 $\pm$ 0.04

I Extreme-Personality Study Configuration

The extreme-personality study runs on the 22 ACI-Bench cases and isolates each HEXACO axis by varying one component of θ at a time with all other axes held at the lowest level and CEFR fixed to B. To compare with PatientSim under the most charitable conditions for that framework, we pair each PWP configuration with the closest behavioral analogue available in the PatientSim parameter space, summarized in Table 10. The pairing reflects that PatientSim does not separate Honesty-Humility from recall and does not expose Conscientiousness or Openness as distinct axes; we select the PatientSim configuration whose dominant behavior most closely overlaps the targeted HEXACO axis.

Table 10: Pairing of extreme PWP configurations with the closest available PatientSim configuration.

Persona	PWP θ	PatientSim parameterization
Dishonest	(3, 1, 1, 1, 1, B)	B / distrust / recall=low / dazed=normal
Emotional	(1, 3, 1, 1, 1, B)	B / overanxious / recall=high / dazed=normal
Extraverted	(1, 1, 3, 1, 1, B)	B / verbose / recall=high / dazed=normal
Frustrated/Skeptical	(1, 1, 1, 3, 1, B)	B / impatient / recall=high / dazed=normal
Disorganized	(1, 1, 1, 1, 3, B)	B / plain / recall=low / dazed=high
Cautious	(1, 1, 1, 1, 1, 3, B)	B / distrust / recall=high / dazed=normal

Conversational model, meta-model, and judge are identical to the realism study (Appendix F).

J Clinician Study Extended Results

Figure 10 reports the per-axis realism subscores (symptom realism, information control, style realism) underlying the binary classification of Section 5.1. AgentClinic falls behind on all three axes, whereas PatientSim and PWP sit close to each other and to Human Rephrase; on information control PatientSim is even slightly ahead of PWP. This does not contradict the information-disclosure results in Section 5.1: the subscore is the rater’s *subjective perception* of how naturally information is volunteered and paced, not a measurement of the underlying disclosure dynamics, and the binary realism classification is a holistic judgment that can be driven by factors beyond these three axes. Table 2 reports how close the human raters and the autorater were to the real parametrization of the personality axes. Figure 11 reports inter-rater agreement for the realism task. With at most a handful of transcripts per rater per source, both the subscore means and the agreement statistics carry substantial between-rater variance attributable to the small sample size per source per rater. The pairwise Cohen’s κ values are near zero on average, reflecting that the binary real-vs-simulated judgment is inherently ambiguous, rather than indicating annotator inconsistency, since even recorded human encounters were classified as real only 52.5% of the time. Aggregating over raters and transcripts recovers a stable signal, and the close agreement between clinicians and the LLM autorater on the personality task (see below) provides an independent check on rater behavior.

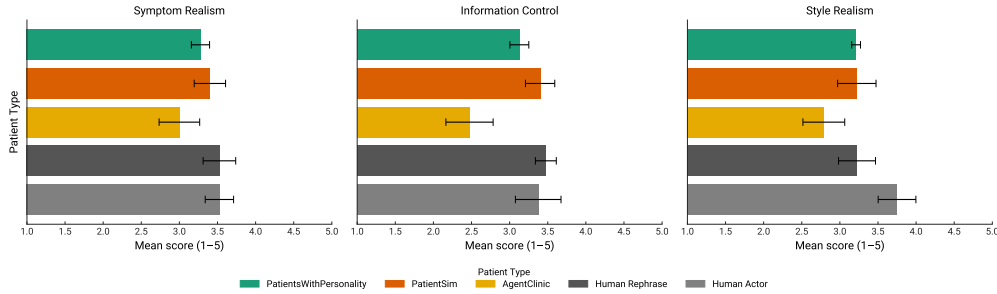
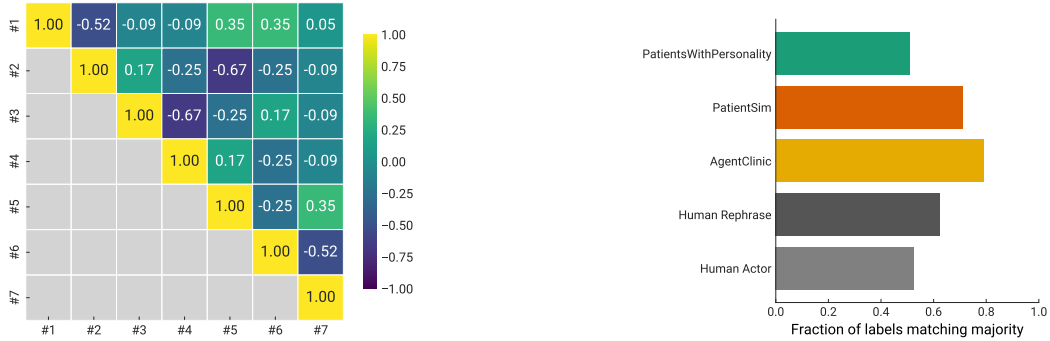


Figure 10: Per-axis realism subscores from the clinician realism task. Mean ratings on a 1–5 scale with SEM error bars across raters.



(a) Pairwise Cohen’s κ between the seven raters on the binary real/simulated label.

(b) Fraction of each rater’s labels matching the majority label of the remaining raters.

Figure 11: Inter-doctor agreement for the realism task.

For each extreme-personality configuration, we average the per-axis HEXACO ratings over conversations and over annotators, separately for the clinicians and for the LLM autorater. Figure 6 plots the autorater mean against the human mean for every (configuration, axis) pair, with the dashed identity line marking perfect agreement. Two summary statistics quantify the alignment: Pearson’s r measures the linear correlation between the two raters’ means, and the intraclass correlation coefficient ICC(2, 1) additionally penalizes systematic offsets and scale differences (a perfect rater pair scores $r = \text{ICC} = 1$).